Reviewer 2: Anonymous

1. In SC-ASC co-culture……..it is not explicitly clear that SC emerged from ASC! Did they also aquire S100 and bipolarity ? (page 2)
2. Incidence of acute….rising constantly. ……. need sentence restrcturing, re-write (page 3)
3. Delete; this. Sentence; and shows that (page 3)
4. addition of growing.. or addition of “growth” (page 3)
5. add “also” here. Sentence; but they also undergo (page 3)
6. in great “interest” replace by 'need'. Sentence; in great need (page 4)
7. It is also esential ….factors. sentence not clear, rewrite (page 4)
8. They could be responsible or partly. “responsible”may be deleted (page 5)
9. It decreases…replace with “it's decrease” (page 5)
10. They secret growing factors…. what are they? (page 5)
11. Some of these Stem cells, …reconsider using 'comma' here (page 5)
12. BMSC is 'Bone marrow stem cells' or 'Bone marrow- derived mesenchymal stem cells' need to define it. (page 5)
13. Some of these stem cells…..stem cell. Whole sentence could be rewritten! Some of these stem cells such as mesenchymal stem cells and bone marrow-derive stem cells have been thorouly used in regenrative medicine. please avoid the adjectives and adverbs like very, quite, well, through out manuscript. (page 5)
14. and are raising their importance… Cells themselves cannot 'raise their importance' replace 'due to' by 'because' (page 5)
15. proved to show…. replace by 'shown' (page 5)
16. Among growing factors ….. its already stated above... avoid repeats (page 6)
17. Introduction is too long, may be cut short to two pages at max. (page 6)
18. “and compares SC”.. highlited portion my be deleted (page 6)
19. Materials and methods … Too much details and elaboration is not required for the standard procedures. Methods could be made more consized, brief and crisp. A lot of the materials not mentioned where from they were obtained etc. (page 6)
20. “animal is” replace by “animal was” (page 6)
21. Each primary antibody was allowed to act… ?? incubated with (page 9)
22. 12 fields of each plate…..was it random selection? (page 9)
23. In plate seeded with SC….The plates where SC were seeded with media (containing 10% FBS) with neuregulin (page 10)
24. “It was obtained a mixture of cells” replace with “A mixture of cells obtained....” (page 10)
25. It is noted that …. 'was' and 'is' in the same sentneces? shaded portion “is” could be deleted (page 10)
26. were classified as SC. It would be better if percentage value of such cell (SC) in the mixture of cells provided. Quantification data needed. (page 10)
27. number of SCs increased slowly.. percent value of the real number shloud be given here! (page 10)
28. In plate seeded with SC.. The plates where SC were seeded with media (containing 10% FBS) supplemented with neuregulin (page 10)
29. positive to anti-S100…replace “to” with “for” (page 10)
30. However, their performance…was higher… However, their proliferation rate was higher.. (page 10)
31. exceeded number of cells. by what factor... in terms of fold, or percentage or absolute numbers.... Please provide a Bar diagram, which could make it more apt. (page 10)

32. between 70% and 100%... is it accurate value or randome estimate? Such data are not provided for 48 hours paltes, why? (page 10)
33. positive to S100…replace “to” with “for” (page 11)
34. The purity of these cultures….(Fig. 2D).. 1. Break into smaller sentences.2. Show it as bar diagram of pei chart 3.in place of range give acurate value. (page 11)
35. As well, … delete (page 11)
36. was already describes…and B).. A comparative data in provided in table 1 could be refered to here as well. (page 11)
37. observed several types of cells… insert 'that' here ..observed that several types of …..(page 11)
38. These cells were positive…as ASC. it is better to say that x% cells stained for S100 and y% cells stained for CD105 and were classified as SC and ASC, respectively. (page 11)
39. These cells were positive to S100, so were classified as SC. if they were S100 positive and SC how come they are different? (page 11)
40. A forth cell type.. third cell type (page 11)
41. As growing time… delete growing (page 11)
42. After 48 hours….B, C and D). This whole para is repitition of above para, it is better that the values were given at the appropriate place in the above para itself. Merge these two para and rewrite this section. (page 11-12)
43. At 48 to 96 hours of culture.….00.1). These data can be provided with the above para where ever relevant. As such this is repetition, and one could connect the values with description when they are given simulteneously at the same place!! (page 12)
44. (Table 2). There is no Table 2!! (page 12)
45. Discussion. discussion is poorly written, whole thing need to be rewritten in unambigous way so that reader could easily grasp what is being said. Also it is almost the results and not discussion. Here authors have present fold changes which were presented in results section as percent changes. The leterature citation to explain the finding is at minimal which indicates that it is mostly the repitition of results. MUST REWRITE (page 12)
46. In current scenario..site. What is the basis of this staement? did authors measure any of the chemicals and presented such data here in this manuscript or other wise? (page 12)
47. important to recognize…this direction. I ma afraid this study does not intend to do as claimed here. At best it is identifying, vaguely, the role of neuregulin in transdifferentiation of ASC into SC like cells, but it fails to support the same with the given set of experiments. (page 13)
48. It was employed adult animals. Replace with 'adult animals were used' (page 13)
49. concerning embryonic stem cell.. insert 'the use of'…concerning the use of embryonic stem cell…(page 13)
50. study is considerably …'was' …study was considerably (page 13)
51. The discussion is marred with "bad' english' as several instances of gramatical errors seen. Have the manuscript proofread by a competent and native english speaker/writer (page 13)
52. shows…showed (page 13)
53. is…was (page 13)
54. 96 h…some where it is hours, at other place h, be consistent (page 13)
55. increases…increased (page 13)
56. they do no longer.,…96 h of culture. ASC lost vimentin expression, although they showed similar proliferation rate as 48 hours (page 13)
57. Further calculation…consider using comma..Further, calculation (page 13)

58. faciliates…facilitated (page 13)
59. is..was (page 13)
60. (1.85) when …but the value is 3.88 at 48 hours with neuregulin!! see the previous sentence. (page 13)
61. (Table 2). where is table 2?? (page 13)
62. results confirmed culture time as.. this is inappropriate conclusion! time is not the factor at all! At best it could be concluded that ' ASC responded similarily in both alone- as well as co-cultures', which is in fact given at the end of this para. (page 13)
63. Regarding question…at same time. too big a sentence, reorganise and break into at least 3 sentences. (page 14)
64. they have already initiated…need to explain what dedifferentiation refers to here? Are not ASC already mesenchymal cells, what would then they dedifferentiate into...? (page 14)
65. These latter…Use 'latter' when referring to the second of two persons but here it is not the case. 'The same authors' suits better. (page 14)
66. therefore this transdifferentiation…do authors intend to use dedifferentiation and transdifferentiation synonymously? if Yes it should be avoided! (page 14)

Figures and tables

67. figure1 ..these cells.. delete these cells (page 19)
68. (red or green;….. I wonder why? why would one use two different secondary antibodies in same set of experiment? (page 19)
69. Figure 2…(red or green;…. But there is no red marker in figure 2!! (page 19)
70. Figure 1…why does the same protein S100 show different kind of stains, red and green, in two images. (page 20)
71. Figure 2….in fig 2, there is no red, but has been mention in the figure legend, red and green!! (page 20)
72. Figure 2A…How would one differentiate SC from ASC in this image? if there was no staining of cells with anti-vimentin at 96 hours, how did authors conclude that vimentine expression decreased by 96 hours? see discussion para 3. (page 20)
73. Figure 2B…if magnification of B image is same 20X why they look so small as compared to other images of same magnification? (page 20)
74. Figure 3…the order and arrangement of figure B and C changed here unlike in figure 1 and 2? (page 20)
75. Table 1….the data could be presented in a better tabular manner. Table is invisible!!

This and other mistake suggest that authors were not serious about the work and manuscript itself. (page 21)

76. Where is table 2? (page 21)